2nd Yr Week 1 Module Assignment

Program1:Triangle Pattern

```c
#include <stdio.h>
#include <conio.h>
/*TRIANGLE*/
void main(){
	int i,j;
	for(i=1;i<=5;i++){
		printf("\n");
		for(j=1;j<=i;j++){
			printf("*");
		}
	}
	getch();

}
```

OUTPUT

```
*
**
***
****
*****
```

Program2:Inverted Triangle

```c
#include <stdio.h>
#include <conio.h>
/*Inverted Triangle*/
void main(){
	int i,j;
	for(i=5;i>=1;i--){
		printf("\n");
		for(j=1;j<=i;j++){
			printf("*");
		}
	}
	getch();

}
```

OUTPUT

```
*****
****
***
**
*
```

Program3:Half Diamond

```
#include<stdio.h>
#include<conio.h>
/*Half Diamond*/
void main(){
	int i,j,n;
	printf("\nEnter the Limit");
	scanf("%d",&n);
	for(i=1;i<=n;i++){
		for(j=1;j<=i;j++){
			printf("*");
		}
		printf("\n");
	}
	for(i=i-2;i>=1;i--){
		for(j=1;j<=i;j++){
			printf("*");
		}
		printf("\n");
	}
	getch();

}
```

OUTPUT

```
*
**
***
****
***
**
*
```

Program4:Diamond

```c
#include<stdio.h>
#include<conio.h>
/*Diamond*/
void main(){
        int i,j;
        for(i=1;i<=5;i++){
                printf("\n");
        for(j=5;j>i;j--){
                printf(" ");
        }
        for(j=1;j<=i;j++){
                printf("*");
        }
        for(j=i-1;j>=1;j--){
                printf("*");
        }
     }
     for(i=i-2;i>=1;i--){
        printf("\n");
        for(j=5;j>i;j--){
                printf(" ");
        }
        for(j=1;j<=i;j++){
                printf("*");
        }
        for(j=i-1;j>=1;j--){
                printf("*");
        }
     }
}
```

OUTPUT

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

Program5:Hollow Square

```c
#include<stdio.h>
#include<conio.h>
#include <stdio.h>
/*Hollow Square*/
int main() {
    int n, i, j;

    printf("Enter size: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++) {
        for(j = 1; j <= n; j++) {
            if(i == 1 || i == n || j == 1 || j == n)
                printf("* ");
            else
                printf("  ");
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT

```
* * * *
*     *
*     *
* * * *
```

Program6:Square

```c
#include<stdio.h>
#include<conio.h>
#include <stdio.h>
#include <stdio.h>
/*SQUARE*/
int main() {
    int n, i, j;

    printf("Enter size: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++) {
        for(j = 1; j <= n; j++) {
            printf("* ");
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

Program7:Pyramid

```c
#include<stdio.h>
#include<conio.h>
#include <stdio.h>
/*PYRAMID*/
int main() {
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++) {

        // Spaces
        for(j = 1; j <= n - i; j++) {
            printf(" ");
        }

        // Stars
        for(j = 1; j <= (2 * i - 1); j++) {
            printf("*");
        }

        printf("\n");
    }

    return 0;
}
```

OUTPUT

```
    *
   ***
  *****
 *******
*********
```

Program8:Inverted Pyramid

```c
#include<stdio.h>
#include<conio.h>
#include <stdio.h>
/*INVERTED PYRAMID*/
int main() {
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = n; i >= 1; i--) {

        // Spaces
        for(j = 1; j <= n - i; j++) {
            printf(" ");
        }

        // Stars
        for(j = 1; j <= (2 * i - 1); j++) {
            printf("*");
        }

        printf("\n");
    }

    return 0;
}
```

OUTPUT

```
*********
 *******
  *****
   ***
    *
```